

SHREYA VONTELA

Tel: +1 (510)-988-1801 | Email : shreya_vontela@mfe.berkeley.edu | [LinkedIn](#) | [GitHub](#) | [Substack](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Principal Data Scientist & AI Engineer with 7+ years of experience architecting high-impact AI systems and quantitative risk frameworks in financial sector. Expert in bridging **Quantitative Risk (PD/LGD, Prepayment, Fraud, Payments)** with **Agentic AI orchestration (RAG, LangChain)** to drive profitability and operational scale. Proven track record of leading cross-functional teams to build end-to-end production pipelines for risk-informed pricing, regulatory automation, and intelligent loss mitigation, resulting in **\$5M+ revenue impact** and **80% reduction in manual oversight**. Adept at translating complex data into strategic, revenue-impacting decisions for global banking portfolios.

EXPERIENCE

Capital One (Discover Financial Services), Principal Data Scientist

Jan 2023 – Oct 2025

Strategic Risk & Portfolio Analytics (Risk/Pricing/Fraud)

- Enterprise Regulatory Monitoring Hub: Led the end-to-end development of a centralized monitoring engine (Snowflake/Airflow) to automate evaluation of 230+ compliance controls. Managed a team of four Data Scientists to scale the framework across Home, Personal, and Student Loan portfolios, reducing manual monitoring effort by 80% and saving \$5M+ in expenses
- Risk-Informed Pricing Optimization: Formulated a causal inference framework to optimize risk-adjusted profitability for home equity products. Designed quasi-experimental pricing variations to estimate elasticity, driving a **~15% increase in funded loan volume**.
- Next-Generation Underwriting & PD Modeling: Engineered and operationalized a high-precision underwriting framework using LightGBM and SHAP-based interpretability to identify non-linear risk drivers. By integrating this model into an automated decisioning engine, replaced legacy manual reviews for low-risk segments, reducing the application-to-funding timeline by 20% and driving a measurable increase in funded loan volume.
- Application Fraud Detection: Built a mortgage fraud-risk scoring framework using application and entity-linkage features; combined anomaly detection, peer outlier scoring, rule overlays, and graph-risk signals to reduce false-positive reviews by 20% while maintaining comparable fraud-capture rates.

Applied AI & LLM Systems

- Agentic Loss Mitigation System: Architected a multi-agent RAG workflow using LangChain to automate borrower case synthesis and modification strategy drafting. Reduced analyst handling time by 45% while ensuring adherence to compliant communication standards.
- Regulatory Copilots & Text-to-SQL: Engineered Gen AI copilots (Snowflake/LangChain) to interpret complex regulatory controls & enable natural language querying for business users, reducing analysis time by 50%.
- Automated Servicing NLP: Developed an LLM-based categorization engine to route borrower complaints and potential compliance risks to specialized servicing teams, cutting manual review time by 70%.

Aspen Capital, Principal Data Scientist

June 2022 – Jan 2023

- Architected a risk-aware iBuyer pricing framework using LightGBM quantile regression to estimate Q10/Q50/Q90 valuations and calibrate prediction intervals for downside stress-testing, enabling the conversion of valuation uncertainty into dynamic BUY offer spreads and the implementation of a SELL-side feedback loop that adjusted future buy offers based on realized resale performance, reducing downside overpay risk by 17%.

PennyMac, VP of Data Science (Underwriting and Credit Risk)

May 2021 – May 2022

- Predictive Credit Risk Modeling: Developed an end-to-end framework of LightGBM classification models using real-time credit variables to predict loan disposition outcomes, generating ~\$25M in revenue
- Deep Learning for Prepayment/Default: Built a neural network on 1TB+ datasets to predict conditional probabilities of prepayment and delinquency. Integrated macroeconomic estimators (Moody's) to manage interest rate and credit risk for a large-scale mortgage portfolio.

JPMorgan Chase & Co, Quant Researcher (Credit Card Payment Fraud)

April 2019 – March 2020

- Built a real-time credit card payment fraud detection system for payment authorization transactions using Kafka-based streaming pipelines and XGBoost models, where low-latency behavioral features such as

transaction velocity, device anomalies, merchant/geolocation deviations, spending-pattern drift, and linked-entity graph signals were computed in real time to generate fraud risk scores within milliseconds and drive approve/decline/challenge decisions while reducing false positives and improving fraud capture.

CalPERS, Investment Associate Intern

Sept 2020 – Nov 2020

- Built a [linear programming model](#) to design an index future roll strategy that minimizes the roll cost. [\[Code\]](#)
- Calculated incremental financing costs for stressed liquidity portfolio by comparing collateral REPO & OIS rates.

Goldman Sachs, Asset Management Intern

April 2018 – June 2018

- Interned as a pension fund advisor for a \$2B AUM client in a multi-asset class macro investment strategy team
- Built [excel based tool](#) to spot investment opportunities across 26 country indices using earnings & sales values

Bank of America Merrill Lynch, Developer, [Best employee Award](#)

July 2015 – June 2017

- Developed a framework for real-time trade data settlement & reconciliation of global Equity & Forex derivatives
- Automated the reconciliation of end of day transactions of banks nostro accounts ([Saved 300+ man hours](#))

FREELANCE CONSULTING PROJECTS

AI-Assisted Endoscopic Guidance System [\[Blog\]](#)

Nov 2025 - Current

- Led development of an AI-assisted endoscopic guidance system combining a [YOLOv8-based anatomical detection model](#) and a CNN “True North” orientation classifier to label throat structures and guide scope positioning in real time; achieved mAP 0.78, 92 % orientation accuracy and optimized models for mobile device deployment.

Trading strategy for tourism industry in USA, Data Science Team at Kayrros [\[Code\]](#)

- Built a neural network with adaptive LR & time series cross validation to predict casino stock price (9.6% return)

Academic/ Personal Projects/ Blogs

- Performed [Geospatial analysis](#) & built a house price prediction model to find top investable zips in USA [\[Code\]](#)
- Modeled the probability of default of [Lending Club loans](#) and calibrated the XGBoost model [\[Code\]](#)
- Performed [PCA on 130 features & used XG Boost Classifier](#) to model trading for Jane St Kaggle Project [\[Code\]](#)
- Built a [convex portfolio optimization model](#) to perform asset allocation based on Sharpe ratio ([87% return](#))

EDUCATION

University of California, Berkeley – Haas School of Business

2020 – 2021

Master of Financial Engineering – Specialization in Machine Learning

Berkeley, CA, USA

Indian Institute of Management, Lucknow, India ([Institute Ranker](#))

2017 – 2019

MBA: Master of Business Administration (Exchange at EMLYON Business School, Lyon)

India and France

Birla Institute of Technology and Science (BITS), Pilani, India

2011 – 2015

Bachelor of Technology in Electronics and Instrumentation Engineering

India

CORE SKILLS

Machine Learning and AI:

Regression & Classification · Ensemble Methods (Random Forest, XGBoost, Bagging, Boosting) · SVM · KNN · Clustering · CNNs & RNNs · Deep Learning · Time Series Forecasting · A/B Testing · NLP (Text Classification, Tokenization, Embeddings, Transformers) · Generative AI (RAG, Prompt Engineering, Agentic AI Systems)

Programming and Tools:

Python · TensorFlow · Scikit-learn · PyTorch · Amazon SageMaker (AWS) · SQL · Git · Linux · Excel · Tableau

Certifications:

Deep Learning (Coursera), TensorFlow in practice (Coursera), R programming (Udemy)

Domain Expertise:

Credit Risk Modeling · Loan Portfolio Analytics · Fraud Detection · Loss Mitigation Modeling · Regulatory Compliance Automation